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BRAKE, BRAKE SYSTEM AND VEHICLE

Abstract

A brake includes: a hydraulic cylinder including a brake hole, and the brake hole including a piston assembly configured to move along an axial direction of the brake hole; a limiting support fixed to the hydraulic cylinder and covering the brake hole; a push rod passing through the limiting support, connected to the piston assembly, and configured to drive the piston assembly to move along the axial direction of the brake hole; and an elastic element disposed on the limiting support and opposite to the piston assembly, and configured to abut against the piston assembly for limiting the piston assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation application of International Patent Application No. PCT/CN2023/105038, filed on Jun. 30, 2023, which is based on and claims priority to and benefits of Chinese Patent Application No. 202211613491.8, filed on Dec. 15, 2022. The entire content of all of the above-referenced applications is incorporated herein by reference.

FIELD

[0002] The present disclosure relates to the field of vehicles, and particularly, to a brake, a brake system, and a vehicle.

BACKGROUND

[0003] In the related art, an existing brake includes a hydraulic cylinder, a piston, and a limiting support. The piston is movably arranged in the hydraulic cylinder. The piston is in transmission connection with a brake pedal of a vehicle. During return of the brake pedal, the piston collides with the limiting support. The piston comes into rigid contact with the limiting support, causing big working noise of the brake during the return of the brake pedal, and generating abnormal sound. In this way, noise, vibration, and harshness (NVH) performance of the entire vehicle is reduced.

SUMMARY

[0004] The present disclosure resolves at least one of technical problems in the related art to some extent.

[0005] Therefore, the present disclosure provides a brake. A piston assembly of the brake abuts against an elastic element during return of a brake pedal. This can reduce collision noise during return of the piston assembly, thereby improving NVH performance of an entire vehicle.

[0006] The present disclosure provides a brake system.

[0007] The present disclosure provides a vehicle.

[0008] The brake according to the present disclosure includes: a hydraulic cylinder, a limiting support, a push rod, and an elastic element.

[0009] The hydraulic cylinder has a brake hole. The brake hole has a piston assembly configured for moving along an axial direction of the brake hole.

[0010] The limiting support is fixed to the hydraulic cylinder and covers the brake hole.

[0011] The push rod passes through the limiting support and is connected to the piston assembly. The push rod is configured to drive the piston assembly to move along the axial direction of the brake hole.

[0012] The elastic element is arranged on the limiting support and opposite to the piston assembly. The elastic element is configured to abut against the piston assembly for limiting the piston assembly.

[0013] According to the brake in the present disclosure, the elastic element is arranged, and the piston assembly abuts against the elastic element during return of a brake pedal of a vehicle, to avoid direct collision between the piston assembly and the limiting support. This can reduce collision noise during return of the piston assembly, thereby improving NVH performance of the entire vehicle.

[0014] The brake system according to the present disclosure includes the foregoing brake.

[0015] According to the brake system in the present disclosure, collision noise during return of a

piston can be reduced, thereby reducing working noise of the brake system, and improving NVH performance of an entire vehicle.

[0016] The vehicle according to the present disclosure includes the foregoing brake system.

[0017] According to the vehicle in the present disclosure, the foregoing brake system is arranged, thereby improving NVH performance of an entire vehicle.

[0018] Other aspects and advantages of the present disclosure will be given in the following description, some of which will become apparent from the following description or may be learned from practices of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 is an exploded view of a hydraulic cylinder, a push rod, a piston, an elastic element and a limiting support according to an embodiment of the present disclosure;

[0020] FIG. 2 is a cross-sectional view of a limiting support according to an embodiment of the present disclosure;

[0021] FIG. 3 is a schematic diagram of an elastic element according to an embodiment of the present disclosure; and

[0022] FIG. 4 is a schematic diagram of an elastic element according to an embodiment of the present disclosure from another angle.

DETAILED DESCRIPTION

[0023] The following describes embodiments of the present disclosure in detail. Examples of the embodiments are shown in the accompanying drawings, and same or similar reference signs in all the accompanying drawings indicate same or similar components or components having same or similar functions. The embodiments described below with reference to the accompanying drawings are examples and merely used for explaining the present disclosure, and should not be construed as a limitation on the present disclosure.

[0024] Referring to FIG. 1 to FIG. 4, a brake **100** according to the embodiments of the present disclosure is described below. The brake **100** may be mounted on a vehicle. However, the present disclosure is not limited thereto. The brake **100** may also be arranged/disposed on another device requiring the brake **100**. The present disclosure is described by using the brake **100** arranged on the vehicle as an example.

[0025] As shown in FIG. 1 to FIG. 4, the brake **100** according to the embodiments of the present disclosure includes: a hydraulic cylinder **10**, a limiting support **20**, a push rod **30**, and an elastic element **40**. The hydraulic cylinder **10** has a brake hole **11**. The brake hole **11** is provided with a piston assembly **60** capable of moving along an axial direction of the brake hole **11**, which may also be understood as that the piston assembly **60** may move relative to the hydraulic cylinder **10** along the axial direction of the brake hole **11**. A part of a structure of the piston assembly **60** is mounted in the brake hole **11**. The limiting support **20** is fixed to the hydraulic cylinder **10** and covers the brake hole **11**. It needs to be noted that the limiting support **20** is arranged outside the hydraulic cylinder **10**. The limiting support **20** may be fixedly mounted on the hydraulic cylinder **10** through a threaded fastener (for example, a bolt or a screw). The limiting support **20** may be clamped on the hydraulic cylinder **10**. The push rod **30** passes through the limiting support **20** and is connected to the piston assembly **60**. The push rod **30** is configured to drive the piston assembly **60** to move along the axial direction of the brake hole **11**. It needs to be noted that one end of the push rod **30** may be in direct or indirect transmission connection with a brake pedal of the vehicle, and the other end of the push rod **30** may be in direct or indirect transmission connection with the piston assembly **60**. When the brake pedal is stumped on, the push rod **30** drives the piston assembly **60** to move in the brake hole **11** along a direction away from the limiting support **20**.

During return of the brake pedal, the push rod **30** drives the piston assembly **60** to move in the brake hole **11** along a direction close to the limiting support **20**.

[0026] The elastic element **40** is an elastic member. For example, the elastic element **40** may be a rubber member. However, the present disclosure is not limited thereto. The elastic element **40** may be made of another elastic material. The elastic element **40** is arranged on the limiting support **20**, and the elastic element **40** is arranged opposite to the piston assembly **60**. The elastic element **40** is configured to abut against the piston assembly **60** for limiting same. It needs to be noted that the elastic element **40** may be fixedly mounted on the limiting support **20**. The elastic element **40** and the limiting support **20** may be integrally formed. The elastic element **40** may be detachably mounted on the limiting support **20**. The present disclosure is described by using the elastic element **40** detachably mounted on the limiting support **20** as an example. The elastic element **40** is arranged opposite to the piston assembly **60** along the axial direction of the brake hole **11**.

[0027] When the brake pedal is stumped on, the push rod **30** drives the piston assembly **60** to move in the brake hole **11** along the direction away from the limiting support **20**. During the return of the brake pedal, the push rod **30** drives the piston assembly **60** to move in the brake hole **11** along the direction close to the limiting support **20**. When the piston assembly **60** returns to an original position, the piston assembly **60** abuts against the elastic element **40** for limiting the same. The elastic element **40** stops the piston assembly **60**, to avoid direct collision between the piston assembly **60** and the limiting support **20**. Due to certain elasticity of the elastic element **40**, the elastic element **40** may absorb collision energy. This can reduce collision noise during return of the piston assembly **60**, thereby improving NVH performance of the entire vehicle. Moreover, the brake **100** in the present disclosure has a simple structure and is easy to be assembled.

[0028] Therefore, the elastic element **40** is arranged, and the piston assembly **60** abuts against the elastic element **40** during the return of the brake pedal of the vehicle, to avoid the direct collision between the piston assembly **60** and the limiting support **20**. This can reduce the collision noise during the return of the piston assembly **60**, thereby improving the NVH performance of the entire vehicle.

[0029] In some embodiments of the present disclosure, as shown in FIG. 1, the piston assembly **60** includes a piston **12** and a sleeve **13**. The sleeve **13** is fixed to an end portion of the piston **12** along an axial direction of the piston **12**. The elastic element **40** is configured to abut against the sleeve **13** for limiting same. The brake further includes a support rod hole adjacent to the brake hole **11**. An axis of the support rod hole is parallel with an axis of the brake hole **11**. The brake further includes a support rod **14** fixedly connected to the piston assembly **60**. The support rod **14** is fixedly connected to the sleeve **13**. An axis of the support rod **14** is parallel with an axis of the piston **12**. When the push rod **30** drives the piston assembly **60** to move along the axial direction of the brake hole **11**, the support rod **14** synchronously moves along the axis of the support rod hole. The brake further includes a signal generator (for example, a magnet) and a signal receiver sensing the signal generator. The signal generator is fixedly connected to the support rod **14**. The signal receiver is arranged on the hydraulic cylinder **10**. The signal receiver is configured to detect a displacement signal after the sleeve **13** is separated from the elastic element **40**. During the return of the brake pedal, the push rod **30** drives the piston assembly **60** to move in the brake hole **11** along the direction close to the limiting support **20**. When the piston assembly **60** returns to the original position, the sleeve **13** abuts against the elastic element **40** for limiting the same.

[0030] In some embodiments of the present disclosure, as shown in FIG. 1, the limiting support **20** has an end wall **21** opposite to the piston assembly **60**. The elastic element **40** is arranged on the end wall **21**. The push rod **30** passes through the elastic element **40**. As shown in FIG. 1 and FIG. 2, the limiting support **20** defines a mounting groove **23** open at one end. A bottom wall of the mounting groove **23** is configured as the end wall **21** opposite to the piston assembly **60**. A flange mounting portion **24** is arranged on an end portion of a side wall of the mounting groove **23** away from the end wall **21**. The flange mounting portion **24** is fixedly connected to the hydraulic

cylinder **10**. The flange mounting portion **24** may be mounted on the hydraulic cylinder **10** through a bolt or a screw. The elastic element **40** is mounted on the limiting support **20**. The elastic element **40** is sleeved on an outer side of the push rod **30**, to cause the push rod **30** to pass through the elastic element **40**. The elastic element **40** is arranged on the end wall **21**, and the push rod **30** passes through the elastic element **40**, to cause the push rod **30** to be connected to the piston assembly **60**. This can make arrangement positions of the elastic element **40** and the push rod **30** appropriate, thereby reducing difficulty of manufacturing the brake **100**.

[0031] In some embodiments of the present disclosure, as shown in FIG. 1, the end wall **21** of the limiting support **20** is provided with a first assembly hole **22**. The elastic element **40** is mounted in the first assembly hole **22**. In some embodiments of the present disclosure, the elastic element **40** is sleeved on the first assembly hole **22**, which may also be understood as that the elastic element **40** is arranged in the first assembly hole **22**. The elastic element **40** is provided with a second assembly hole **41**. The push rod **30** passes through the second assembly hole **41**. The first assembly hole **22** passes through the end wall **21** along a thickness direction of the end wall **21** of the limiting support **20**. The second assembly hole **41** passes through the elastic element **40** along an axial direction of the elastic element **40**. A central axis of the second assembly hole **41** is parallel with a central axis of the first assembly hole **22**. Further, the central axis of the second assembly hole **41** is collinear with the central axis of the first assembly hole **22**. Through such an arrangement/configuration, the elastic element **40** can be mounted on the end wall **21** of the limiting support **20**, and the push rod **30** can also pass through the elastic element **40**.

[0032] In some embodiments of the present disclosure, as shown in FIG. 1 to FIG. 3, the elastic element **40** may include: a sleeve body **42** and a first limiting flange **43**. An inner circumferential wall of the sleeve body **42** forms the second assembly hole **41**. The sleeve body **42** is mounted in the first assembly hole **22**. It needs to be noted that the sleeve body **42** is mounted in the first assembly hole **22**, and the sleeve body **42** defines the second assembly hole **41**. The second assembly hole **41** passes through the sleeve body **42** along the axial direction of the elastic element **40**. The first limiting flange **43** is arranged on an outer circumferential wall of the sleeve body **42**. The first limiting flange **43** extends around the sleeve body **42** from the outer circumferential wall of the sleeve body **42**. The first limiting flange **43** is configured to abut against the sleeve **13** of the piston assembly **60** for limiting same. Further, the first limiting flange **43** and the sleeve body **42** may be integrally formed. The first limiting flange **43** is arranged around the sleeve body **42** along a circumferential direction of the sleeve body **42**. During the return of the brake pedal, the push rod **30** drives the piston **12** to move in the brake hole **11** along the direction close to the limiting support **20**. When the piston **12** returns to the original position, the sleeve **13** abuts against the first limiting flange **43**. This can avoid collision between the piston assembly **60** and the limiting support **20**.

[0033] In some embodiments of the present disclosure, as shown in FIG. 3 and FIG. 4, the first limiting flange **43** has a first side surface **431** and a second side surface **432** oppositely arranged along the axial direction of the elastic element **40**. The first side surface **431** abuts against the end wall **21**. The second side surface **432** is configured to abut against the piston assembly **60**. After the elastic element **40** is mounted in the first assembly hole **22**, the first limiting flange **43** is located in the mounting groove **23**. The first side surface **431** of the first limiting flange **43** is arranged opposite to the end wall **21**. The first side surface **431** of the first limiting flange **43** abuts against an inner surface of the end wall **21**. The second side surface **432** of the first limiting flange **43** is arranged opposite to the piston assembly **60**. During the return of the brake pedal, the push rod **30** drives the piston **12** to move in the brake hole **11** along the direction close to the limiting support **20**. When the piston **12** returns to the original position, the piston assembly **60** abuts against the second side surface **432**, to cause the piston assembly **60** to come into surface contact with the elastic element **40**. This can ensure a contact area between the piston assembly **60** and the elastic element **40**, so that the elastic element **40** can better limit the piston **12**. In this way, the collision

between the piston assembly **60** and the limiting support **20** can be further avoided, thereby better improving the NVH performance of the entire vehicle.

[0034] In some embodiments of the present disclosure, a limiting structure configured to increase a roughness of the second side surface **432** is arranged on the second side surface **432**. The limiting structure may be arranged as at least one of a printed groove structure recessed toward the inside of the first limiting flange **43** and a printed protruding structure protruding from the first limiting flange **43**. The limiting structure is arranged on the second side surface **432**, so that the roughness of the second side surface **432** can be increased. In this way, when the piston assembly **60** comes into contact with the second side surface **432**, friction of a contact surface between the second side surface **432** and the piston assembly **60** can be increased, thereby avoiding slippage between the piston assembly **60** and the elastic element **40**.

[0035] In some embodiments of the present disclosure, as shown in FIG. 3 and FIG. 4, at least one clearance groove **433** may be arranged on the second side surface **432**. The quantity of clearance grooves **433** arranged on the second side surface **432** is one or more. The clearance groove **433** is arranged on the second side surface **432**, so that a contact area between the elastic element **40** and the piston assembly **60** can be reduced, and adhesion between the elastic element **40** and the piston assembly **60** can be relieved. In this way, when the piston assembly **60** collides with the elastic element **40**, collision noise during the return of the piston assembly **60** can be further reduced, thereby further improving the NVH performance of the entire vehicle.

[0036] In some embodiments of the present disclosure, as shown in FIG. 3 and FIG. 4, multiple clearance groove **433** may be provided. The multiple clearance grooves **433** are successively arranged at intervals along a circumferential direction of the first limiting flange **43**, to form a boss **44** between two adjacent clearance grooves **433**. Further, the multiple clearance grooves **433** are successively and evenly arranged at intervals along the circumferential direction of the first limiting flange **43**. A spacing between any two adjacent clearance grooves **433** is the same. The multiple clearance grooves **433** are arranged on the second side surface **432**, and the boss **44** is formed between the two adjacent clearance grooves **433**. In this way, when the piston assembly **60** collides with the first limiting flange **43**, the boss **44** comes into contact with the piston assembly **60**, so that the contact area between the elastic element **40** and the piston assembly **60** can be further reduced, and the adhesion between the elastic element **40** and the piston assembly **60** can be better relieved. In this way, when the piston assembly **60** collides with the elastic element **40**, the collision noise during the return of the piston assembly **60** can be further reduced, thereby further improving the NVH performance of the entire vehicle.

[0037] In some embodiments of the present disclosure, as shown in FIG. 4, each of bosses **44** has a first surface **441** and a second surface **442** oppositely arranged along the circumferential direction of the first limiting flange **43**. An angle β is formed between the first surface **441** and the second surface **442** of at least one boss **44**. Further, the angle β is formed between the first surface **441** and the second surface **442** of each of the bosses **44**, and $0^\circ < \beta$. Through such an arrangement, structural strength of the boss **44** can be enhanced, so that a deformation amount of the elastic element **40** can be reduced. However, the present disclosure is not limited thereto. The first surface **441** and the second surface **442** of the boss **44** may be parallel to each other, and in this case, β is 0° .

[0038] In some embodiments of the present disclosure, as shown in FIG. 2 and FIG. 3, the first assembly hole **22** has a first hole section **221** and a second hole section **222** connected to each other, and the first hole section **221** is located at an end of the second hole section **222** close to the hydraulic cylinder **10**. As shown in FIG. 2, the first hole section **221** is located at an end of the second hole section **222** close to the mounting groove **23**. A diameter of the first hole section **221** is greater than a diameter of the second hole section **222**. The first limiting flange **43** is assembled in the first hole section **221**. A shape of the first limiting flange **43** matches with a shape of the first hole section **221**. For example, when the first limiting flange **43** is circular, the first hole section **221** is a circular hole. An outer diameter of the first limiting flange **43** is less than or equal to the

diameter of the first hole section **221**. Through such an arrangement, it can be ensured that the first limiting flange **43** is assembled in the first hole section **221**.

[0039] Further, after the first limiting flange **43** is assembled in the first hole section **221**, the first limiting flange **43** may have an interference fit with the first hole section **221**, so that the elastic element **40** can be reliably assembled in the first assembly hole **22**.

[0040] In some embodiments of the present disclosure, as shown in FIG. 2 and FIG. 3, a thickness A of the first limiting flange **43** and a depth B of the first hole section **221** satisfy a relational expression: $B \leq A$. After the first limiting flange **43** is assembled in the first hole section **221**, through such an arrangement, the first limiting flange **43** can be parallel with the inner surface of the end wall **21**, or the first limiting flange **43** can protrude from the inner surface of the end wall **21**. In this way, when the piston **12** returns to the original position, it can be ensured that the piston **12** does not collide with the limiting support **20**.

[0041] In some embodiments of the present disclosure, as shown in FIG. 1 to FIG. 3, the elastic element **40** may further include a second limiting flange **45**. The second limiting flange **45** is arranged on the outer circumferential wall of the sleeve body **42**. The second limiting flange **45** extends around the sleeve body **42** from the outer circumferential wall of the sleeve body **42**. The second limiting flange **45** extends around the sleeve body **42** along the circumferential direction of the sleeve body **42**. The first limiting flange **43** and the second limiting flange **45** are spaced away from each other along the axial direction of the elastic element **40**. The first limiting flange **43** and the second limiting flange **45** are respectively arranged at two axial ends of the elastic element **40**. After the elastic element **40** is mounted on the limiting support **20**, the second limiting flange **45** is located outside the limiting support **20**, and the second limiting flange **45** abuts against an outer wall opposite to the end wall **21**. The second limiting flange **45** abuts against an outer surface of the end wall **21**. The first limiting flange **43** is located in the mounting groove **23**, the first limiting flange **43** abuts against the inner surface of the end wall **21**, and the second limiting flange **45** abuts against the outer surface of the end wall **21**. In this way, the elastic element **40** can be prevented from moving out of the first assembly hole **22**, so that the elastic element **40** can be tightly assembled on the limiting support **20**, thereby preventing the limiting support **20** from being separated from the elastic element **40**.

[0042] In some embodiments of the present disclosure, an outer surface of the elastic element **40** may be provided with a non-stick coating. The non-stick coating may be arranged as a Teflon coating. The non-stick coating is configured to relieve adhesion between the elastic element **40** and a metal member (for example, the piston **12**). In this way, when the piston assembly **60** collides with the elastic element **40**, the collision noise during the return of the piston assembly **60** can be further reduced, thereby further improving the NVH performance of the entire vehicle.

[0043] In some embodiments of the present disclosure, the elastic element **40** may withstand a temperature range from at least -40°C . to 120°C . Optional materials for manufacturing the elastic element **40** may include silica gel, ethylene propylene diene monomer, and the like.

[0044] In some embodiments of the present disclosure, a central axis of the limiting support **20** is parallel with a central axis of the piston assembly **60**. For example, the central axis of the limiting support **20** coincides with the central axis of the piston assembly **60**. Further, the central axis of the piston assembly **60** is parallel with the central axis of the brake hole **11**. For example, the central axis of the piston assembly **60** coincides with the central axis of the brake hole **11**. Through such an arrangement, the piston assembly **60** and the push rod **30** are easily assembled, and a structure of the brake **100** is simplified, thereby facilitating production and manufacturing of the brake **100**.

[0045] Other compositions of the brake **100** according to the embodiments of the present disclosure, such as the piston **12**, are known to those of ordinary skill in the related art and are not described in detail herein.

[0046] A brake system according to an embodiment of the present disclosure is provided. The brake system may be an electrically controlled brake system or a wire-controlled brake system. The brake

system includes the brake **100** in the foregoing embodiment. This can reduce collision noise during return of a piston **12**, and reduces working noise of the brake system, thereby improving NVH performance of an entire vehicle.

[0047] A vehicle according to an embodiment of the present disclosure is provided. The vehicle includes the brake system in the foregoing embodiments, thereby improving NVH performance of the entire vehicle.

[0048] In the description of the present disclosure, the description of the reference terms such as “an embodiment”, “some embodiments”, “exemplary embodiments”, “example”, “specific example”, or “some examples” means that the features, structures, materials, or characteristics described with reference to the embodiment or example are included in at least one embodiment or example of the present disclosure. In the present disclosure, schematic descriptions of the foregoing terms do not necessarily refer to the same embodiment or example. In addition, the described features, structures, materials, or characteristics may be combined in any one or more embodiments or examples in an appropriate manner.

[0049] Although the embodiments of the present disclosure have been shown and described, those of ordinary skill in the related art should understand that various changes, modifications, replacements, and variations may be made to the embodiments without departing from the principle and spirit of the present disclosure, and the scope of the present disclosure is defined by the appended claims and equivalents thereof.

Claims

1. A brake, comprising: a hydraulic cylinder comprising a brake hole, and the brake hole comprising a piston assembly configured to move along an axial direction of the brake hole; a limiting support fixed to the hydraulic cylinder and covering the brake hole; a push rod passing through the limiting support, connected to the piston assembly, and configured to drive the piston assembly to move along the axial direction of the brake hole; and an elastic element disposed on the limiting support and opposite to the piston assembly, and configured to abut against the piston assembly for limiting the piston assembly.
2. The brake according to claim 1, wherein the limiting support has an end wall opposite to the piston assembly, the elastic element is disposed on the end wall, and the push rod passes through the elastic element.
3. The brake according to claim 2, wherein the end wall comprises a first assembly hole, the elastic element is mounted in the first assembly hole and comprises a second assembly hole, and the push rod passes through the second assembly hole.
4. The brake according to claim 3, wherein the elastic element comprises a sleeve body and a first limiting flange, an inner circumferential wall of the sleeve body forms the second assembly hole, the first limiting flange is disposed on an outer circumferential wall of the sleeve body and around the sleeve body, and the first limiting flange is configured to abut against the piston assembly for limiting the piston assembly.
5. The brake according to claim 4, wherein the elastic element is sleeved on the first assembly hole.
6. The brake according to claim 4, wherein the first limiting flange has a first side surface and a second side surface oppositely disposed along an axial direction of the elastic element, the first side surface abuts against the end wall, and the second side surface is configured to abut against the piston assembly.
7. The brake according to claim 6, wherein a limiting structure configured to increase a roughness of the second side surface is disposed on the second side surface.
8. The brake according to claim 6, wherein at least one clearance groove is disposed on the second side surface.
9. The brake according to claim 8, wherein a plurality of clearance groove are disposed on the

second side surface at intervals along a circumferential direction of the first limiting flange, and a boss is formed between two adjacent clearance grooves.

10. The brake according to claim 9, wherein the boss has a first surface and a second surface oppositely disposed along the circumferential direction of the first limiting flange, an angle β is formed between the first surface and the second surface of the boss.

11. The brake according to claim 4, wherein the first assembly hole has a first hole section and a second hole section connected to each other, the first hole section is located at an end of the second hole section close to the hydraulic cylinder, a diameter of the first hole section is greater than a diameter of the second hole section, and the first limiting flange is disposed in the first hole section.

12. The brake according to claim 11, wherein a thickness A of the first limiting flange and a depth B of the first hole section satisfy $B \leq A$.

13. The brake according to claim 4, wherein the elastic element further comprises a second limiting flange disposed on the outer circumferential wall of the sleeve body and around the sleeve body, the first limiting flange and the second limiting flange are spaced away from each other along the axial direction of the elastic element, and the second limiting flange is located outside the limiting support and abuts against an outer wall of the limiting support opposite to the end wall.

14. The brake according to claim 1, wherein an outer surface of the elastic element comprises a non-stick coating.

15. The brake according to claim 1, wherein the piston assembly comprises a piston and a sleeve fixed to an end portion of the piston along an axial direction of the piston, and the elastic element is configured to abut against the sleeve for limiting the sleeve.

16. The brake according to claim 15, further comprising a support rod hole, a support rod, a signal generator, and a signal receiver, wherein: the support rod hole is adjacent to the brake hole, and an axis of the support rod hole is parallel with an axis of the brake hole; the support rod is connected to the piston assembly and the sleeve, an axis of the support rod is parallel with an axis of the piston, and the support rod synchronously moves along the axis of the support rod hole when the push rod drives the piston assembly to move along the axial direction of the brake hole; and the signal receiver is configured to sense the signal generator, the signal generator is connected to the support rod, the signal receiver is disposed on the hydraulic cylinder and configured to detect a displacement signal after the sleeve is separated from the elastic element.

17. A brake system, comprising a brake, wherein the brake comprises: a hydraulic cylinder comprising a brake hole, and the brake hole comprising a piston assembly configured to move along an axial direction of the brake hole; a limiting support fixed to the hydraulic cylinder and covering the brake hole; a push rod passing through the limiting support, connected to the piston assembly, and configured to drive the piston assembly to move along the axial direction of the brake hole; and an elastic element disposed on the limiting support and opposite to the piston assembly, and configured to abut against the piston assembly for limiting the piston assembly.

18. A vehicle, comprising a brake system, wherein the brake system comprises a brake, and the brake comprises: a hydraulic cylinder comprising a brake hole, and the brake hole comprising a piston assembly configured to move along an axial direction of the brake hole; a limiting support fixed to the hydraulic cylinder and covering the brake hole; a push rod passing through the limiting support, connected to the piston assembly, and configured to drive the piston assembly to move along the axial direction of the brake hole; and an elastic element disposed on the limiting support and opposite to the piston assembly, and configured to abut against the piston assembly for limiting the piston assembly.
